A Mid-Semester Report On

Lattice Based Post-Quantum Cryptography

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Part I
NTRU Cryptosystem – Meghraj Goswami

1 Introduction

The rapid advancement of quantum computing poses a threat to modern cryptographic infrastructure. Shor's algorithm demonstrates that a quantum computer can factor large integers and compute discrete logarithms in polynomial time—$O((\log n)^3)$—completely breaking RSA and Elliptic Curve Cryptography (ECC) at any practical key size in a short time frame [1].

In response, the cryptographic community has shifted towards **Post-Quantum Cryptography (PQC)**, algorithms that remain secure even against quantum adversaries. The **NTRU** cryptosystem (N-th degree Truncated polynomial Ring Unit), introduced by Hoffstein, Pipher, and Silverman in 1996 [2], is one of the oldest and most analysed PQC proposals. Its security rests on the hardness of the Shortest Vector Problem (SVP) in certain lattices, for which no efficient quantum algorithm is known as of yet.

Key advantages of NTRU over RSA and ECC

- No known quantum speedup for the underlying lattice problem.
- Fast polynomial operations.
- Compact keys: public keys are approximately 700 bytes at the 128-bit security level, versus several kilobytes for RSA at equivalent security.
- NTRU has been a prominent fixture in the NIST PQC standardization process.[3]

2 Mathematical Prerequisites

2.1 Groups

Definition 2.1 (Group)**.** A **group** is a pair $(G, \cdot)$ where G is a set and $\cdot : G \times G \to G$ is a binary operation satisfying:

1. **Closure:** $a \cdot b \in G$ for all $a, b \in G$.
2. **Associativity:** $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
3. **Identity:** $\exists\, e \in G$ such that $a \cdot e = e \cdot a = a$.
4. **Inverses:** For each $a \in G$, $\exists\, a^{-1} \in G$ with $a \cdot a^{-1} = e$.

If additionally $a \cdot b = b \cdot a$ for all $a, b \in G$, the group is called **abelian**.

Example 2.1 (Clock arithmetic ($\mathbb{Z}/12\mathbb{Z}$, $+$))**.** Integers modulo 12 under addition: $10 + 4 \equiv 2 \pmod{12}$.
The identity is 0; the inverse of 3 is 9 since $3 + 9 \equiv 0$. This is an abelian group.

2.2 Rings and Fields

Definition 2.2 (Ring)**.** A **ring** is a triplet $(R, +, \cdot)$ where $(R, +)$ is an abelian group, multiplication is associative, and multiplication distributes over addition. A ring is **commutative** if $ab = ba$ for all $a, b \in R$.

A **field** is a commutative ring where every nonzero element has a multiplicative inverse.

Example 2.2. $\mathbb{Z}$, $\mathbb{Z}/n\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$ are rings. $\mathbb{Q}$ and $\mathbb{R}$ are fields; $\mathbb{Z}$ is not (the element 2 has no multiplicative inverse in $\mathbb{Z}$).

2.3 Polynomial Rings

Definition 2.3 (Polynomial Ring)**.** Let R be a commutative ring. The polynomial ring $R[x]$ consists of all formal sums $a_0 + a_1x + \cdots + a_nx^n$ where $a_i \in R$, under coefficient-wise addition and standard polynomial multiplication.

2.4 Convolution Polynomial Rings

NTRU operates in quotient rings of the following form. Fix a positive integer N and define:

$$\mathcal{R} = \frac{\mathbb{Z}[x]}{(x^N - 1)}, \qquad \mathcal{R}_q = \frac{(\mathbb{Z}/q\mathbb{Z})[x]}{(x^N - 1)}, \qquad \mathcal{R}_p = \frac{(\mathbb{Z}/p\mathbb{Z})[x]}{(x^N - 1)}.$$

The ideal $(x^N - 1)$ enforces $x^N = 1$, so all exponents reduce modulo N, and every element can be represented as a polynomial of degree at most $N - 1$. Meanwhile, the coefficients are taken modulo q or p in $\mathcal{R}_q$ and $\mathcal{R}_p$ respectively.

Multiplication in these rings is the **convolution product** $\star$:

$$c(x) = a(x) \star b(x), \quad c_k = \sum_{\substack{i+j \equiv k \\ (\mathrm{mod}\ N)}} a_i b_j.$$

The wrap-around of higher-degree terms is what gives this product a 'cyclic' character.

Example 2.3 (Convolution product with $N = 5$)**.** Let $a(x) = 1 - 2x + 4x^3 - x^4$ and $b(x) = 3 + 4x - 2x^2 + 5x^3 + 2x^4$. Computing coefficient by coefficient using $x^5 = 1$:

$$a(x) \star b(x) = -13 + 20x - 7x^2 + 19x^3 + 5x^4.$$

2.5 Ternary Polynomials

Definition 2.4 (Ternary polynomial set)**.** For non-negative integers d_1, d_2 with $d_1 + d_2 \leq N$,

$$T(d_1, d_2) = \big\{a(x) \in \mathcal{R} \mid \text{exactly } d_1 \text{ coefficients equal } +1,\ d_2 \text{ equal } -1,\ \text{rest } 0\big\}.$$

These sparse polynomials have small coefficients. Private keys are chosen from $T(d_1, d_2)$ precisely because small coefficients make decryption work (see Section 3.5) and make exhaustive search difficult (see Section 5).

2.6 Polynomial Inversion

Key generation requires computing $f(x)^{-1}$ modulo both q and p. A polynomial $F_q(x)$ is an inverse of $f(x)$ in $\mathcal{R}_q$ if

$$f(x) \star F_q(x) \equiv 1 \pmod{q,\ x^N - 1}.$$

When N and q are both prime, the inverse exists if and only if $\gcd(f(x),\ x^N - 1) = 1$ in $(\mathbb{Z}/q\mathbb{Z})[x]$, and can be computed using the **extended Euclidean algorithm** for polynomials. If f has no inverse, a new f is chosen.

3 Cryptosystem

3.1 Public Parameters

The system is set up with public parameters (N, p, q, d) satisfying:

- N is prime.
- $\gcd(p, q) = 1$ and $\gcd(N, q) = 1$.
- $q > (6d + 1)p$ (required for correct decryption; see Theorem 3.1).

3.2 Key Generation

Alice generates her keys as follows:

1. Choose random private polynomials $f(x) \in T(d + 1, d)$ and $g(x) \in T(d, d)$.
2. Compute inverses of f in both quotient rings:

$$F_q(x) = f(x)^{-1} \text{ in } \mathcal{R}_q, \qquad F_p(x) = f(x)^{-1} \text{ in } \mathcal{R}_p.$$

3. Compute the **public key**:

$$h(x) \equiv F_q(x) \star g(x) \pmod{q}.$$

Private key: $(f,\ F_p)$. **Public key:** h.

3.3 Encryption

Bob represents his plaintext as $m(x) \in \mathcal{R}$ with coefficients in $\left[-\frac{p}{2}, \frac{p}{2}\right]$.

1. Choose a fresh random ephemeral polynomial $r(x) \in T(d, d)$.
2. Compute the **ciphertext**:

$$e(x) \equiv p \cdot r(x) \star h(x) + m(x) \pmod{q}.$$

Because $r(x)$ is drawn fresh for each encryption, NTRU is a *probabilistic* scheme: encrypting the same message twice yields completely different ciphertexts.

3.4 Decryption

Alice recovers $m(x)$ from $e(x)$ in three steps:

1. Compute $a(x) \equiv f(x) \star e(x) \pmod{q}$.
2. **Center-lift:** adjust each coefficient of $a(x)$ to its representative in $\big(-q/2,\, q/2\big]$.
3. Recover plaintext: $m(x) \equiv F_p(x) \star a(x) \pmod{p}$.

3.5 Correctness

Theorem 3.1 (Decryption correctness)**.** *If $q > (6d+1)p$, the decryption procedure recovers $m(x)$ exactly.*

Proof. Expanding $a(x)$ and substituting $h(x) = F_q(x) \star g(x)$:

$$\begin{aligned} a(x) &\equiv f(x) \star e(x) \pmod{q} \\ &\equiv f(x) \star \big(p \cdot r(x) \star h(x) + m(x)\big) \pmod{q} \\ &\equiv p \cdot f(x) \star F_q(x) \star g(x) \star r(x) + f(x) \star m(x) \pmod{q} \\ &\equiv p \cdot g(x) \star r(x) + f(x) \star m(x) \pmod{q}, \end{aligned}$$

where the last step uses $f \star F_q \equiv 1 \pmod{q}$.

The coefficients of $A(x) := p \cdot g \star r + f \star m$ in $\mathcal{R}$ (before reduction mod q) are now bounded. Since $g \in T(d,d)$ and $r \in T(d,d)$, each coefficient of $g \star r$ is at most $2d$ in absolute value. Since $f \in T(d+1,d)$ and the coefficients of m are in $\big[-\frac{p}{2}, \frac{p}{2}\big]$, each coefficient of $f \star m$ is at most $(2d+1)\frac{p}{2}$ in absolute value. Therefore:

$$|A_k| \le p \cdot 2d + (2d+1)\frac{p}{2} < (6d+1)\frac{p}{2} < \frac{q}{2}.$$

Since every coefficient of $A(x)$ already lies in $(-q/2, q/2]$, the center-lift recovers $A(x)$ *exactly* in $\mathcal{R}$ with no modular wrap-around:

$$a(x) = p \cdot g(x) \star r(x) + f(x) \star m(x).$$

Reducing modulo p kills the blinding term ($p \cdot g \star r \equiv 0 \pmod{p}$):

$$F_p \star a \equiv F_p \star f \star m \equiv m \pmod{p}. \qquad \square$$

4 Worked Example

The full protocol is illustrated with the toy parameters $(N, p, q, d) = (7, 3, 41, 2)$.

4.1 Key Generation

Alice chooses:

$$f(x) = -1 + x^2 + x^3 - x^4 + x^6 \in T(3,2), \quad g(x) = -x - x^2 + x^4 + x^6 \in T(2,2).$$

Using the extended Euclidean algorithm she computes:

$$F_p(x) = f(x)^{-1} \pmod{3,\ x^7 - 1} = 1 + x + x^2 + x^3 + 2x^5 + x^6,$$
$$F_q(x) = f(x)^{-1} \pmod{41,\ x^7 - 1} = 37 + 2x + 40x^2 + 21x^3 + 31x^4 + 26x^5 + 8x^6.$$

The public key is:

$$h(x) = F_q(x) \star g(x) \pmod{41} = 30 + 26x + 8x^2 + 38x^3 + 2x^4 + 40x^5 + 20x^6.$$

4.2 Encryption

Bob's plaintext and ephemeral key:

$$m(x) = -x^5 + x^3 + x^2 - x + 1, \qquad r(x) = x^6 - x^5 + x - 1.$$

He computes the ciphertext $e(x) \equiv 3 \cdot r(x) \star h(x) + m(x) \pmod{41}$:

$$e(x) = 25 + 3x + 40x^2 + 2x^3 + 4x^4 + 19x^5 + 31x^6.$$

4.3 Decryption

Step 1. Multiply by $f(x)$ mod 41:

$$f(x) \star e(x) \equiv 40 + x + 40x^2 + 40x^3 + 33x^4 + 10x^5 + x^6 \pmod{41}.$$

Step 2. Center-lift to $(-20.5,\ 20.5]$ (e.g. $40 \mapsto -1$, $33 \mapsto -8$):

$$a(x) = -1 + x - x^2 - x^3 - 8x^4 + 10x^5 + x^6.$$

Step 3. Multiply by $F_p(x)$ mod 3:

$$F_p(x) \star a(x) \pmod{3} = -x^5 + x^3 + x^2 - x + 1 = m(x). \quad \checkmark$$

5 Security Analysis

5.1 Key Recovery Problem

Definition 5.1 (NTRU Key Recovery Problem)**.** Given the public key $h(x) \in \mathcal{R}_q$, find ternary polynomials $f(x)$ and $g(x)$ such that $f(x) \star h(x) \equiv g(x) \pmod{q}$.

Since $h = F_q \star g$, knowing f immediately yields $g = f \star h \pmod{q}$ (and vice versa). The difficulty of this problem is equivalent to finding a short vector in the **NTRU lattice**.

5.2 Brute Force and Meet-in-the-Middle

An attacker could exhaustively search over all $f \in T(d_1, d_2)$. The search space has size:

$$\binom{N}{d_1}\binom{N-1}{d_2} = \frac{N!}{d_1!\,d_2!\,(N-d_1-d_2)!}.$$

Note that any of the N cyclic rotations of a valid f also decrypts correctly, so the effective search space is $\#T(d_1, d_2)/N$.

Example 5.1 ($(N, p, q, d) = (251, 3, 257, 83)$)**.** The search space for $f \in T(84, 83)$ is $\approx 2^{389}$ polynomials; accounting for $N = 251$ rotations gives roughly $2^{381.6}$ effective trials. A meet-in-the-middle (birthday) attack takes the square root, requiring $\approx 2^{190.8}$ operations—still completely infeasible.

6 Conclusion

NTRU is a mature, well-studied post-quantum cryptosystem. Its algebraic foundation in truncated polynomial rings provides a clean mathematical structure, and its reliance on lattice hardness gives resilience against both classical and quantum adversaries. Its combination of small key sizes, fast operations, and strong security evidence makes it one of the most practical post-quantum proposals available today.

This section forms part of a group report on Post-Quantum Cryptography. The initial content was developed from a presentation by Meghraj Goswami.

References

[1] P.W. Shor. *Algorithms for Quantum Computation: Discrete Logarithms and Factoring.* In: *Proceedings of the 35th Annual IEEE Symposium on Foundations of Computer Science (FOCS)*, 1994, pp. 124–134. https://doi.org/10.1109/SFCS.1994.365700

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[3] *NIST Post-Quantum Standardization Effort.* https://csrc.nist.gov/projects/post-quantum-cryptography